

The Current State of Sustainability in Music

Posse: Academic Immersion Day Presentation

by Anish Ghosh

6/5/26





Anish Ghosh

Age: 25

Posse: 7, 2018–2022, Syracuse University

Level: 2022 B.S. Music Business (Newhouse), 2026 M.S. Music Engineering (UM)

Honors:

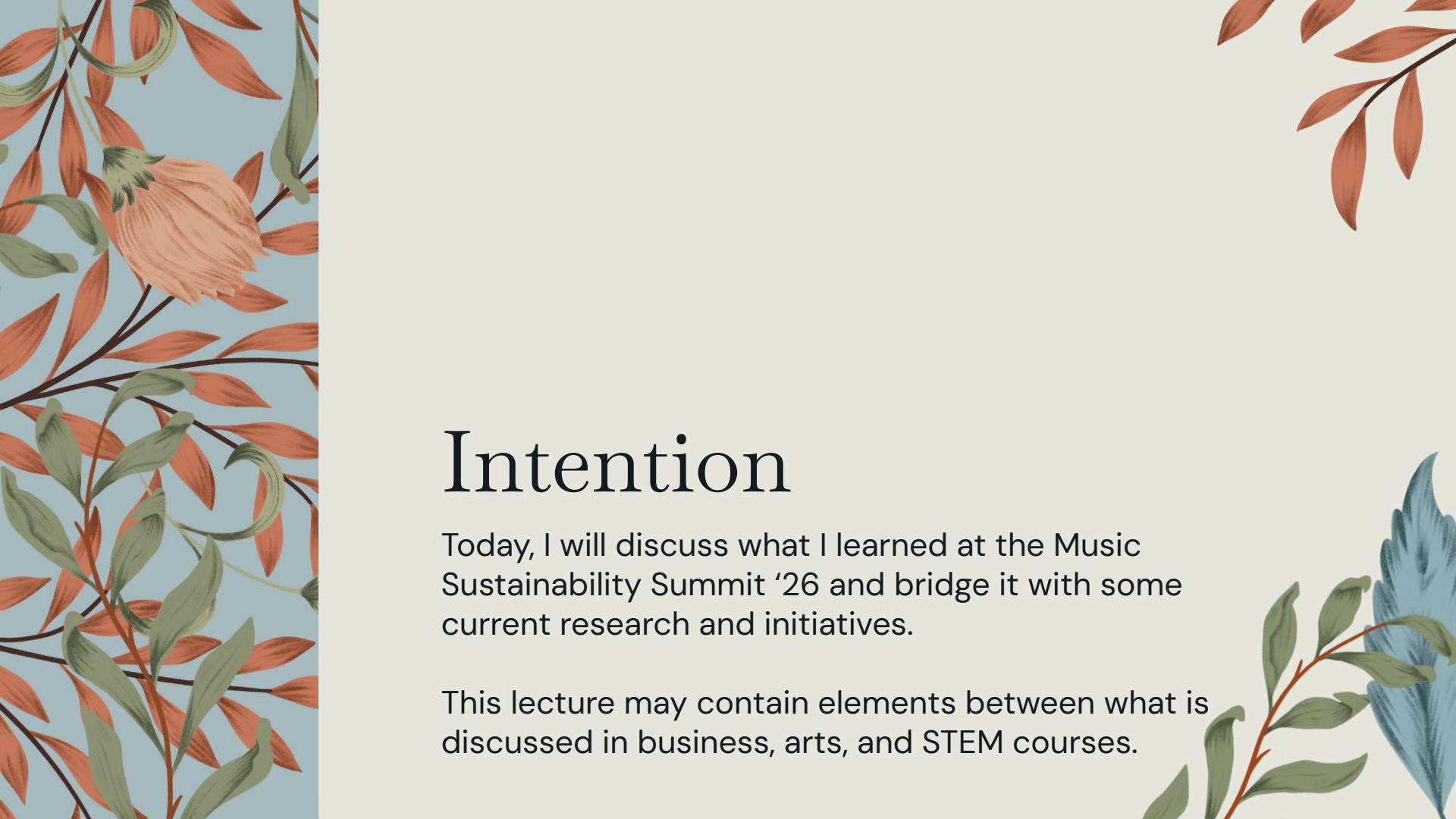
Music Sustainability Summit '26 Scholar

American Popular Music Education '26 Presenter

Audio Engineering Society Europe: Intelligent Audio Student Design Competitor

Being a Posse Scholar, Bengali and proud, having family and friends, and having 2 doggos



The slide features decorative floral illustrations. On the left, a vertical strip shows a branch with orange-red leaves and a large, textured, light-orange flower against a light blue background. In the top right corner, there are more orange-red leaves. In the bottom right corner, there are green leaves and a large, light blue leaf.

Intention

Today, I will discuss what I learned at the Music Sustainability Summit '26 and bridge it with some current research and initiatives.

This lecture may contain elements between what is discussed in business, arts, and STEM courses.

A composite image of Earth from space, showing the Americas and surrounding oceans. The image has a dark, grainy, and somewhat distorted appearance, with some text and numbers overlaid. The text "DOOMSDAY IS C" is prominently displayed in the center. Above it, the year "1991" is visible. At the top, there is a small box containing the text "CH" and "1992/03/31".

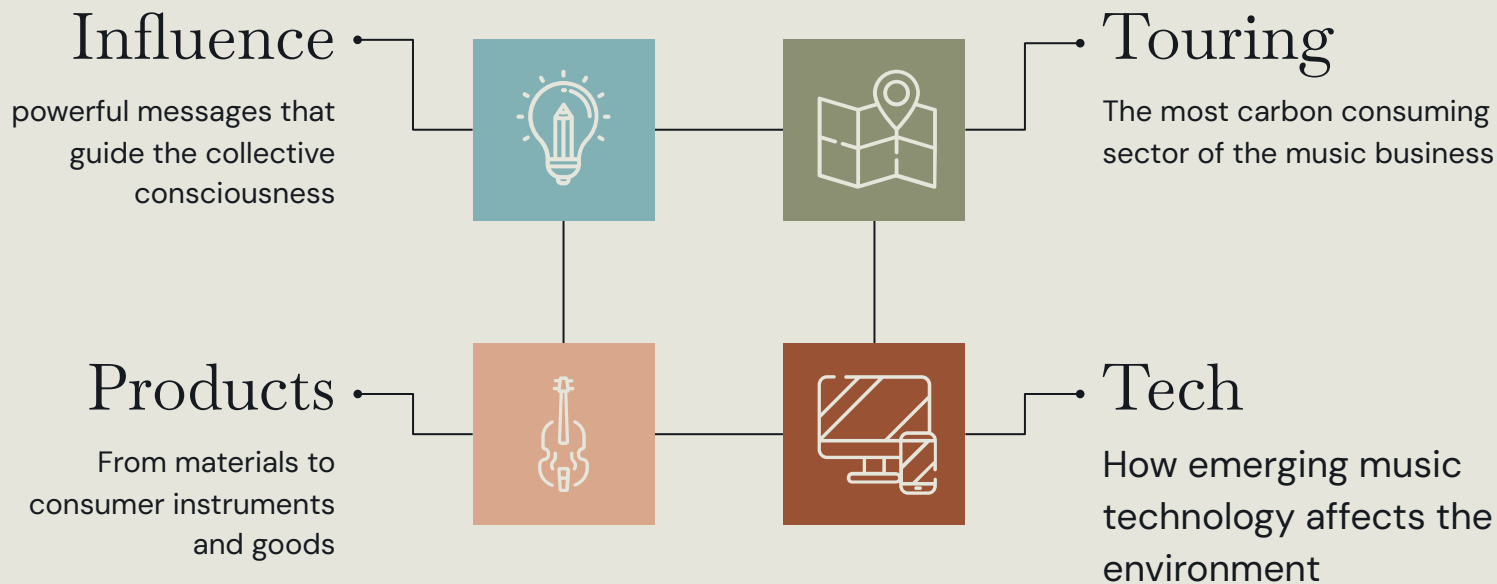
DOOMSDAY IS C

The image features a light beige background with decorative botanical illustrations in the corners. The top-left corner shows a branch with dark green leaves and a single orange-red flower. The top-right corner has a blue leaf. The bottom-left corner features a branch with green leaves. The bottom-right corner includes a blue leaf, a green leafy branch, and a branch with reddish-brown leaves.

Question:

What about music is
beneficial or detrimental
for the environment?

Sections





01

Influence

Music contains powerful messages that guide the collective consciousness



The power of tunes :)

Music, art and media share stories and world views to wide audiences.

The reach of entertainment and communications can start global movements

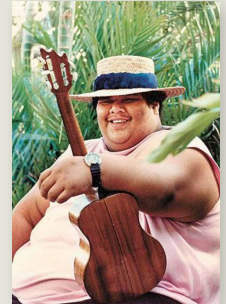
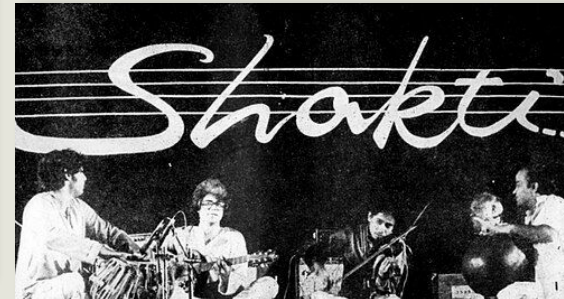
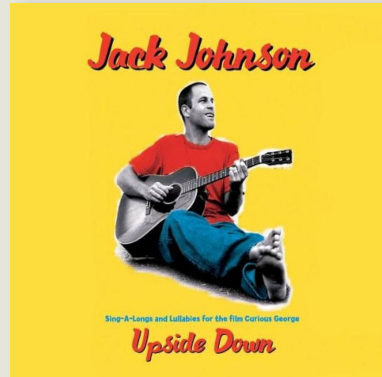
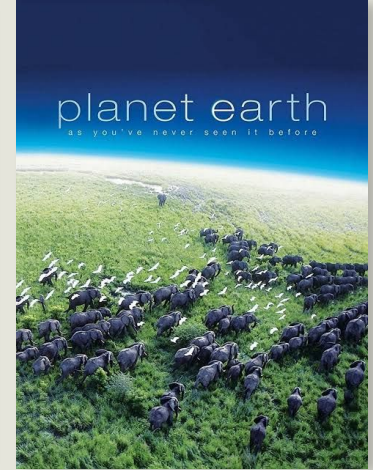
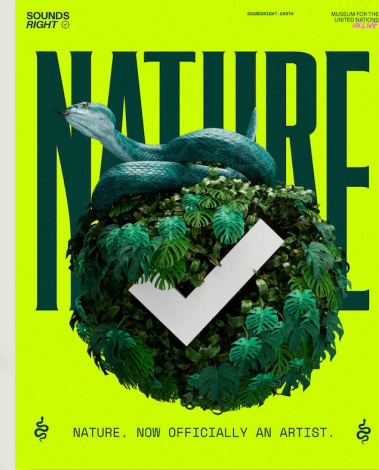


The image features a light beige background with decorative floral and leaf patterns in the corners. In the top-left corner, there are dark green leaves and a single orange-red flower. In the top-right corner, there are blue leaves. In the bottom-left corner, there are dark green leaves. In the bottom-right corner, there are blue and orange-red leaves. The text is centered in the middle of the image.

Question:

What music or media
helps you connect with
the world?

Here's some music and media that connects me to the world:





“No one will protect what they don't
care about, no one will care about
what they have never experienced”

—David Attenborough

Sharing:



World Views

Artists like *Jack Johnson* create music that express their world view on simple living and environmental virtues



Experiences

Artist like *Billie Eilish* create unforgettable experiences for artists that show a lifestyle maybe worth trying



02

Touring

How changes in touring culture has impacted the environment



REVERB'S STORY



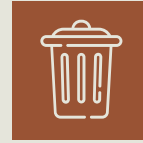
Sustainability in Touring



Energy



Food



Waste





Energy

Companies like *Overdrive Energy Solutions* are

- Working to create dependable electric generators
- Connecting those generators to solar panels or the grid
- Reducing the need for petroleum generators
- Pushing the needle on sustainable energy in entertainment

And this push requires innovations in engineering, promotion in marketing, and a shift in the corporate culture.





Food

Artists like *Billie Eilish* have introduced

- Sourcing food from local farms for events to reduce shipping costs, provide real and healthy foods to fans, and give back to the community that hosts the events
- Plant-based options for fans with dietary restrictions and reducing the overall carbon footprint of the event, it may even promote a lifestyle change to some fans.

IFYKYK: historically, humans had a mainly plant-based diet and eating plant-based can reduce an individual's GHG emissions by up to 75%





Waste

Pollution is a huge issue after a concert or festival!

Events can:

- Promote recycling consciousness at the event
- Promote the use of reusable containers
- And use biodegradable materials

Which leads us to our next topic...



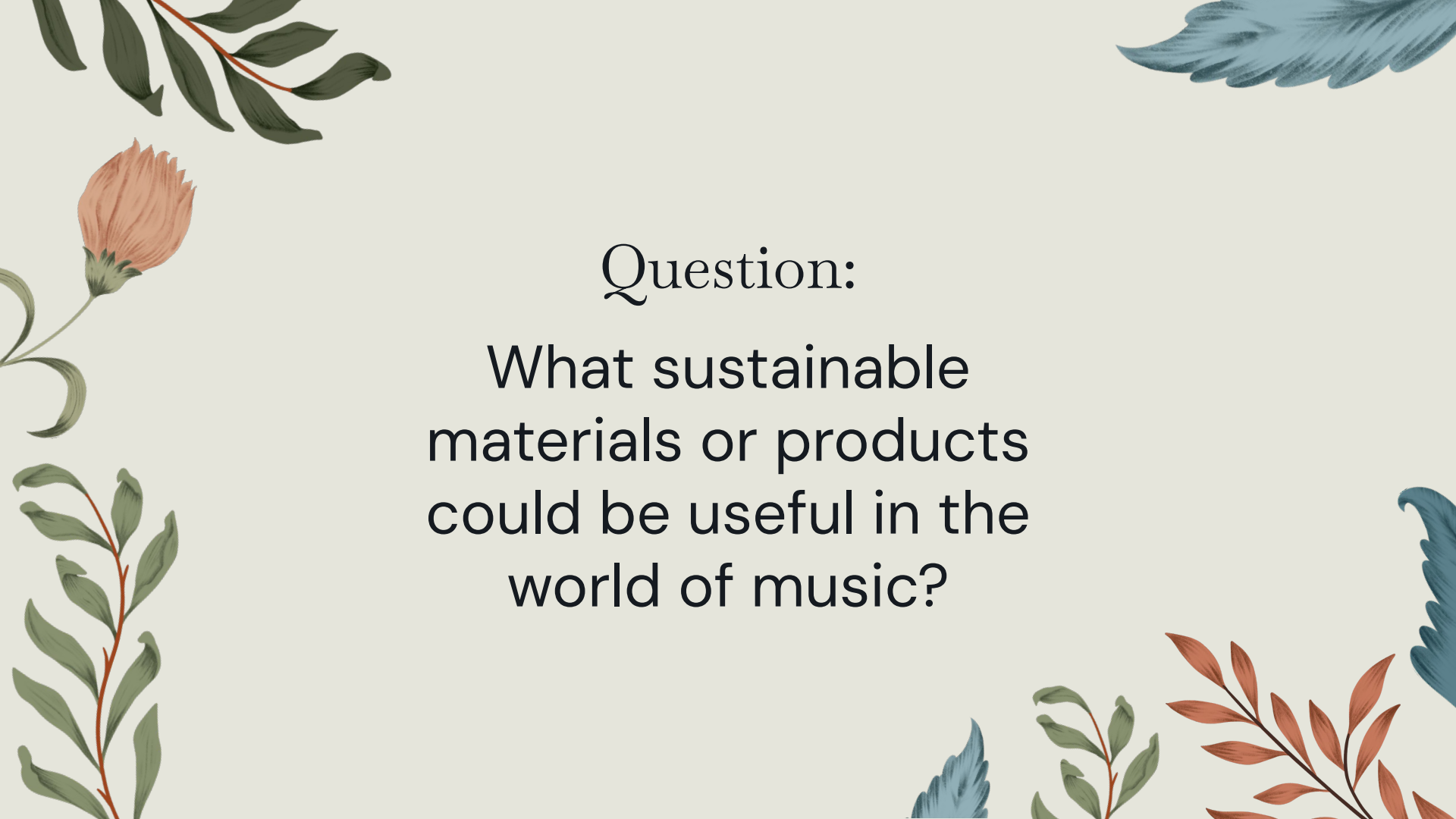


03

Products & Materials

New developments in materials and consumer goods tackle issues at the source.



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Question:

What sustainable
materials or products
could be useful in the
world of music?



Crazy Cool Innovation

Bioplastics

Plastic like materials made from seaweed that can dissolve in water

Carbon Dots

The base material for making LED lights and modern televisions

Bioluminescence

Using natural processes to illuminate items in the dark

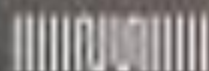
Magents

Require rare earth metals and are needed for infrastructure and mechanical engineering



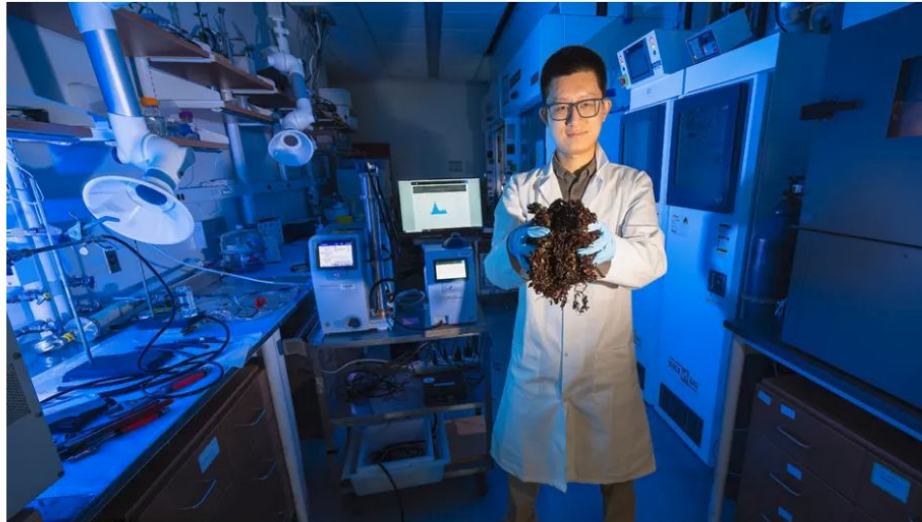


Paige Whitehead
Lux Bio

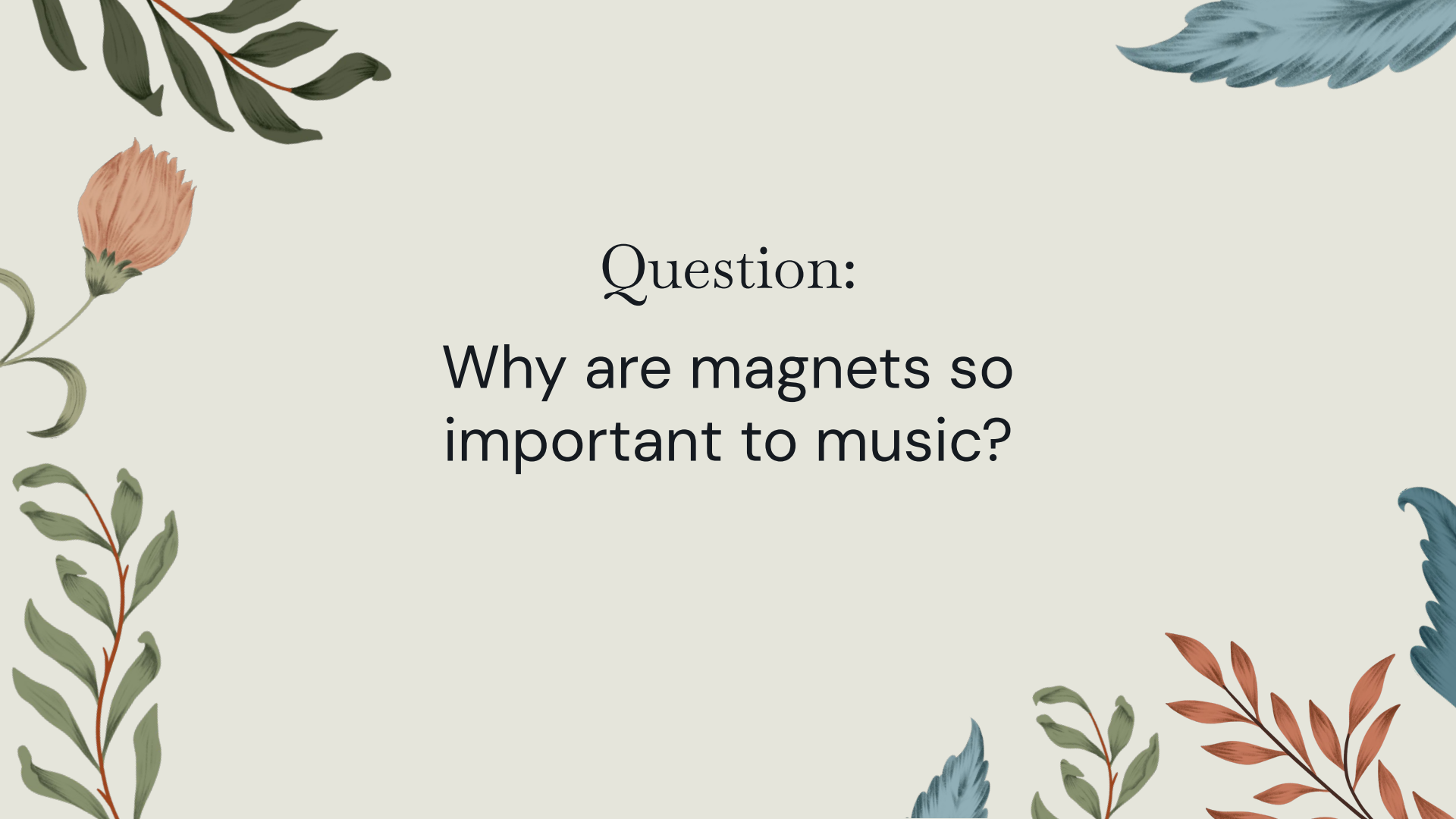


From flatscreens to bioimaging: Putting sargassum seaweed to good use

Viewing sargassum seaweed as a resource rather than a waste, a University of Miami doctoral student is experimenting with the macroalgae to produce fluorescent carbon dots, which show tremendous potential in a multitude of fields.

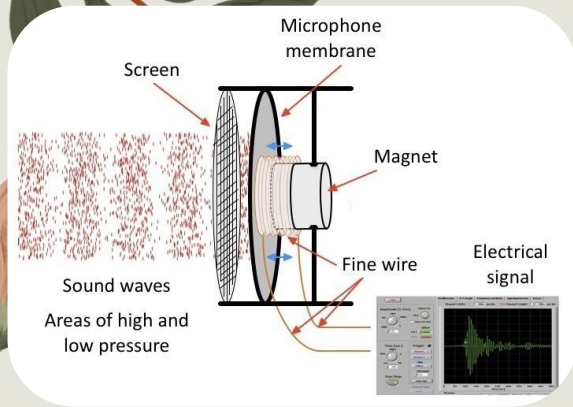


Yiming Xi, a Ph.D. candidate in chemical engineering in the College of Engineering, is researching potential uses for sargassum seaweed. *Photo: Joshua Prezant/University of Miami*

The image features a light beige background with decorative botanical illustrations in the corners. The top-left corner shows a branch with dark green leaves and a single orange-red flower. The top-right corner has a blue, feathery leaf. The bottom-left corner displays a branch with green leaves. The bottom-right corner contains a branch with reddish-brown leaves and a blue, feathery leaf.

Question:

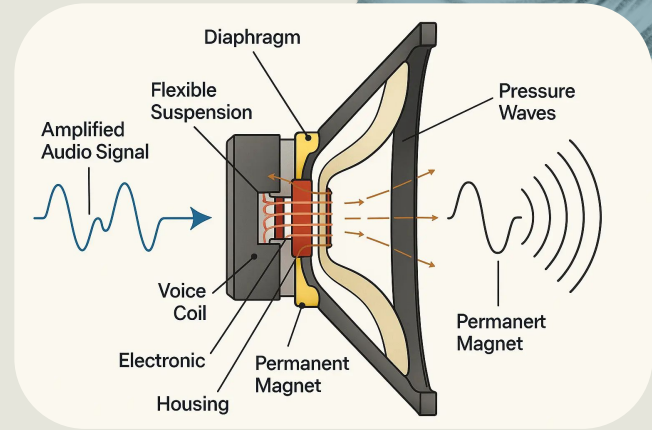
Why are magnets so
important to music?



Answer:

Speakers and Microphones

Niron Magnetics has created Iron Nitride (Fe_{16}N_2) based magnets, one of their main applications will be for commercial speakers and mics.



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Question:

What is one of the most
rare materials in the
universe?

Tip: I learned this from a reel

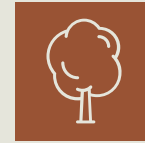
Sustainability in Wood Materials



Martin Guitars

Chose the route of engineering new materials in the lab to replace traditional hardwoods. Food waste, Kombucha Scoby, and Algae can be repurposed to create hardwoods.

Fun fact: guitar enthusiasts were the reason the US opposed bills designed to weaken anti-illegal logging protections, they testified before congress and created a movement called “No More Blood Wood Campaign”



Taylor Guitars

Chose the route of using abundant or sustainable wood species to break traditions like the “solid black ebony” instrument fret board,



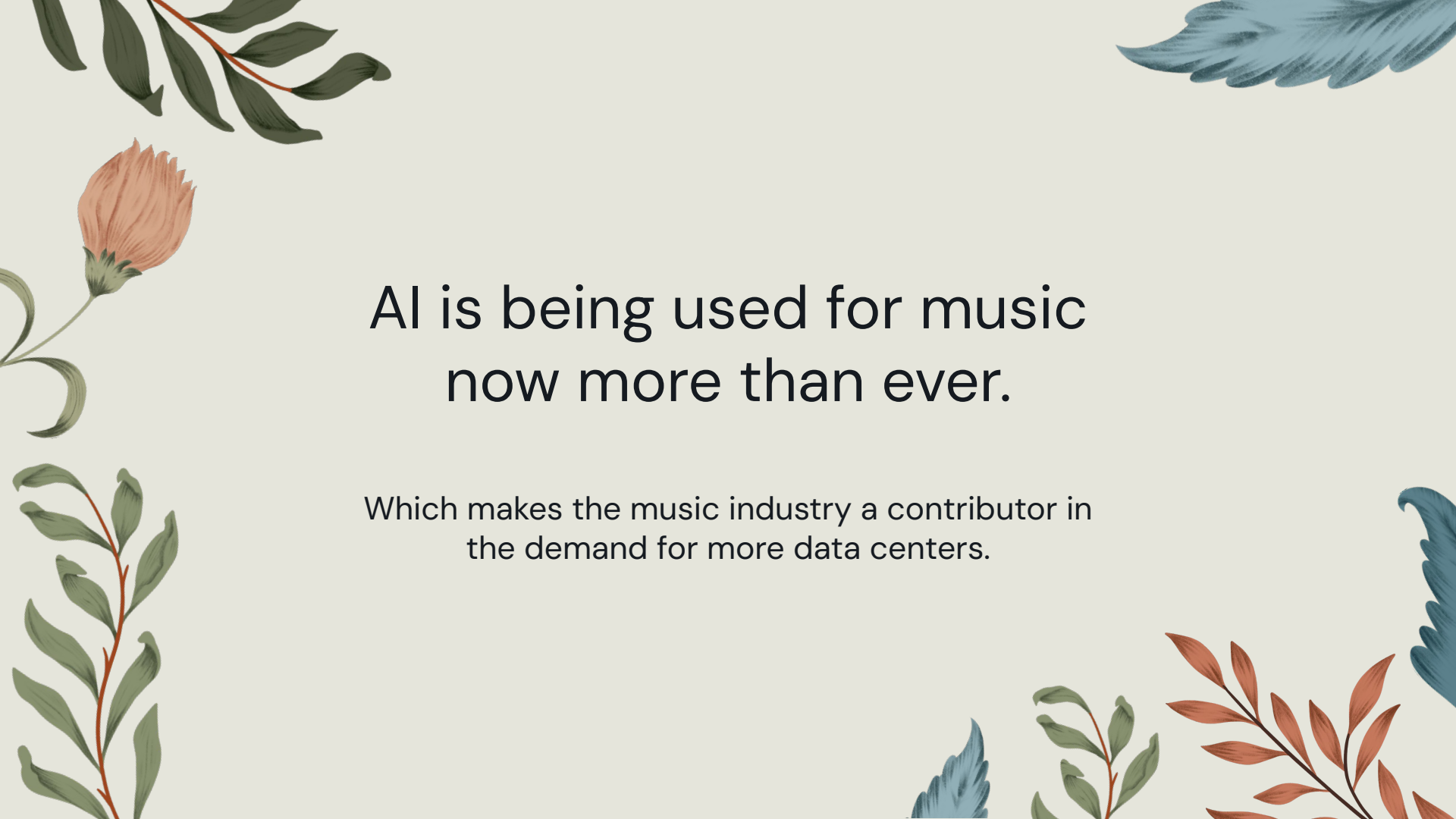


04

Technology

Emerging tech is transforming the concerns and solutions to climate activism.



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AI is being used for music
now more than ever.

Which makes the music industry a contributor in
the demand for more data centers.

Artificial Intelligence and Music



Comp Cost

How much the computer has to process before achieving its end goal

Energy Usage

The immediate segway from computation cost concerns

Water Usage

Water can be used for cooling and cleaning tech infrastructure

Data Centers

Combine the above three concerns with deforestation

Creative Control

Aside from these concerns, creative autonomy is limited with these models

Copyright

Copyright, royalties, and rights to artists' work used in AI is also a concern.



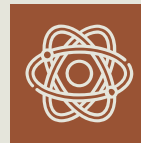
But It's Not All Doom and Gloom



Forest
Restoration



Water
Restoration



Research &
Innovation





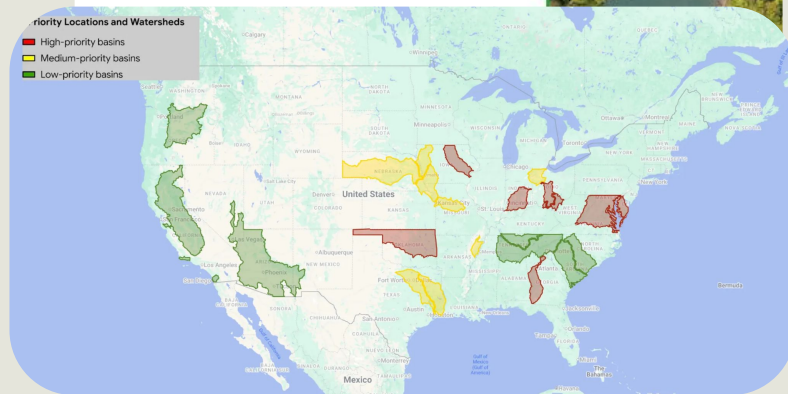
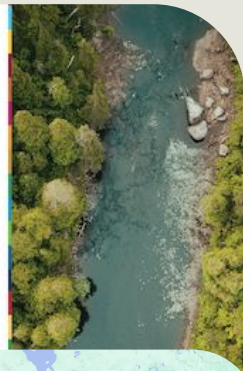
17 YEARS LATER

VWBA Framework

Tech companies have big pockets
Some nonprofits hold these companies
accountable for their damage throughout
the *Volumetric Water Benefit Accounting
Framework*.

Companies like Google use some of their
massive income for restoring rural and
uninhabited water reserves and
infrastructures for reducing their impact
and for better public perception

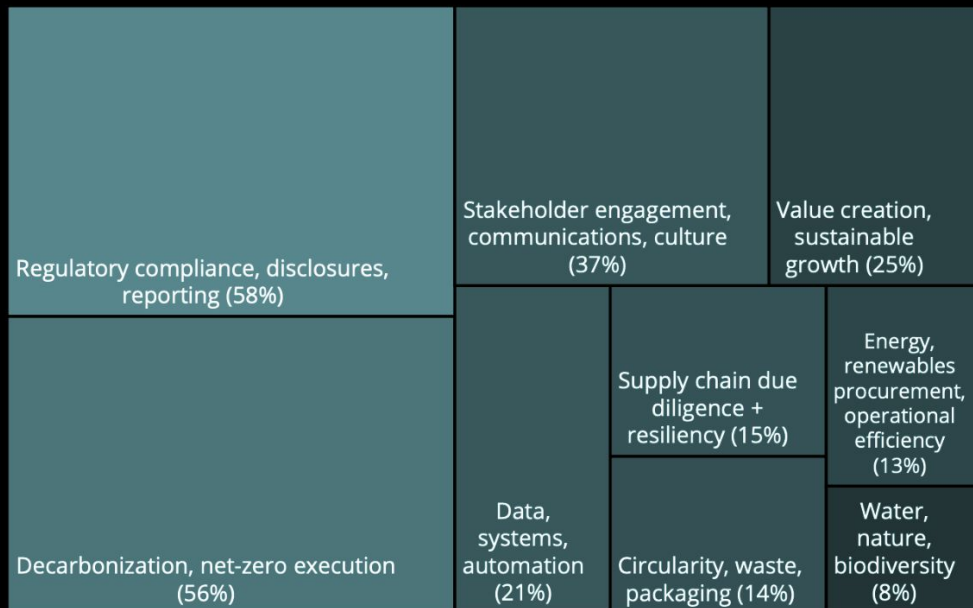
Volumetric Water Benefit
Accounting (VWBA):
**A Practical Guide to
Implementing Water
Replenishment Targets**



Deloitte CSO 2026 Survey

CSOs Priorities across Industries

Box size indicates relative proportion of CSOs that cited each priority area



Insights for the Music Industry

CSOs in consumer, media, & entertainment largely agree with these priorities. However:

- Energy management is a central priority for media and entertainment
- While CSOs in media and entertainment echo the focus on compliance, consumer CSOs often prioritize climate risk and opportunity assessments
- Consumer CSOs have a stronger focus on systems upgrades and automation (i.e., for reporting)



Deloitte CSO 2026 Survey

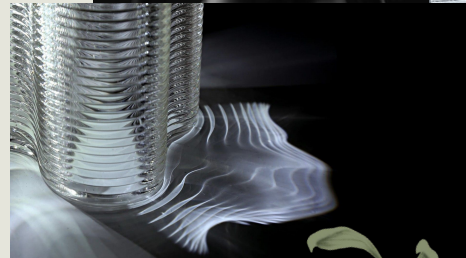
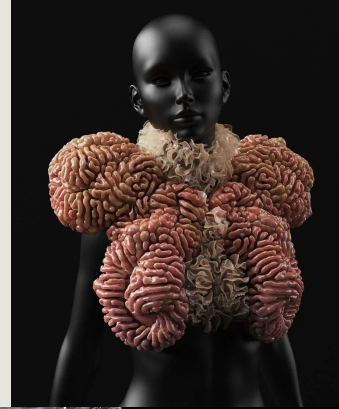
AI	Value Drivers	Capital Investments
What are Sustainability functions using AI for? - To increase sustainability team member efficiency (70%) - For advanced sustainability reporting and data analytics (57%) - To review draft disclosures and assess stakeholder reaction (24%)	Which levers do teams apply to drive value? </	

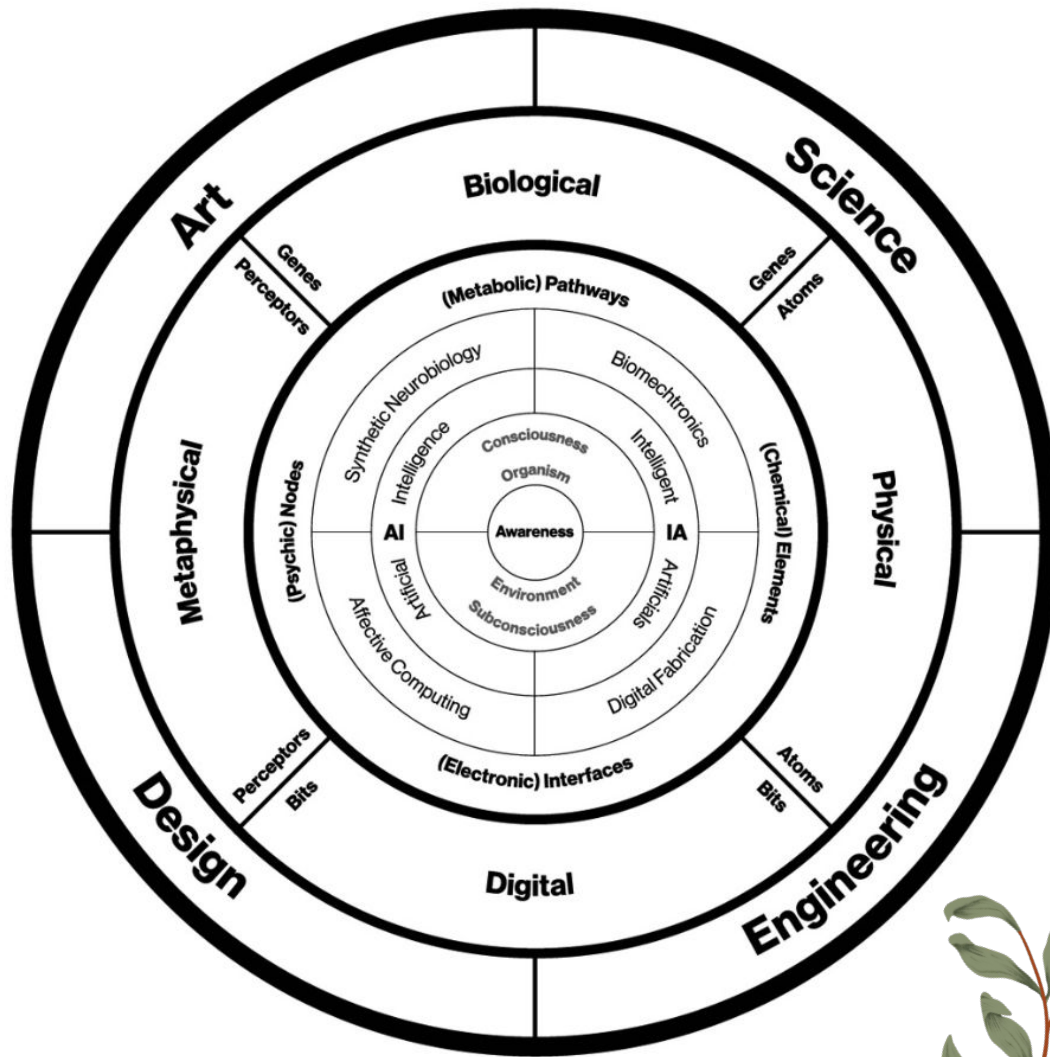
So how does this all fit in the bigger picture?

I'll take a page out of one of my favorite design thinker's books:

Neri Oxman's Material Ecology.

Here's what she thinks about using data and AI for her design process, **you don't have to agree with her**, but she posits some interesting points



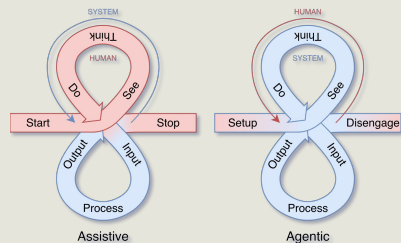


Some of my Relevant Interests

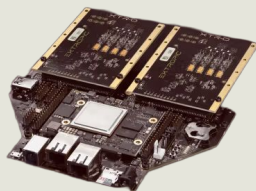
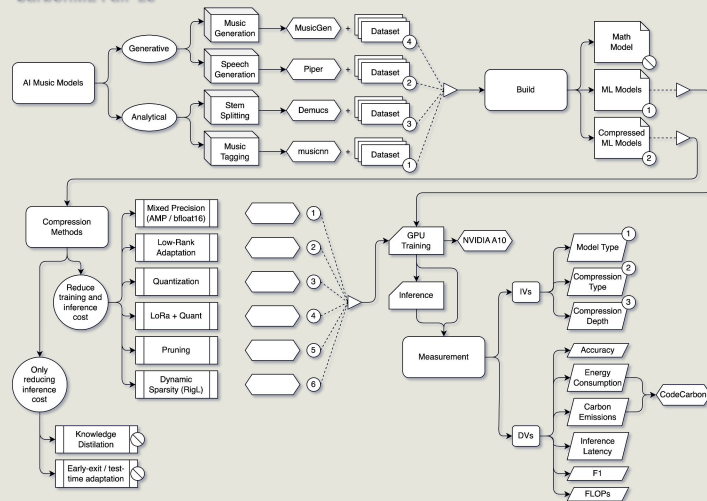
The Impact of Artificially Intelligent Tools on Music Production and Education

Computation Cost and Carbon Emission Reduction of Music ML Models

Thermodynamic Chips for Carbon offsets in Music ML Hardware



CarbonML Fall '25



Not Relevant But Fun

Produced a ton of fun music, logos, graphics, and websites

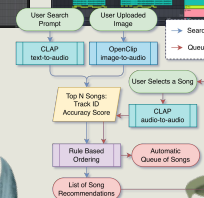
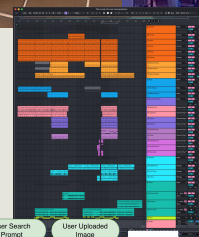
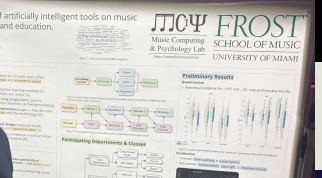
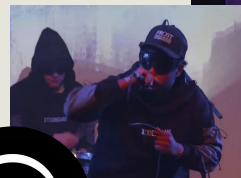
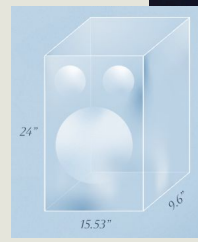
Helped create a reverb plugin that sound like Campus Concert Halls

Using Machine Learning for Diversifying Music Recommendation Algorithms

Designed and built a bass guitar cabinet

Attended and presented at conferences

Performed for a songwriting contest in Amsterdam and performed a metal version of FE!N as a meme at the end of year student concert (I played drums and rapped with autotune on lol)





Advice for navigating college



Interdisciplinary

Draw knowledge from multiple fields

Balance

Challenge yourself but don't overwhelm yourself. Take care of your body and mind.

Resonance

Find what speaks to your heart and soul

Mentors

Stay connected with mentors that inspire you. Make sure you communicate what you want to learn/do.

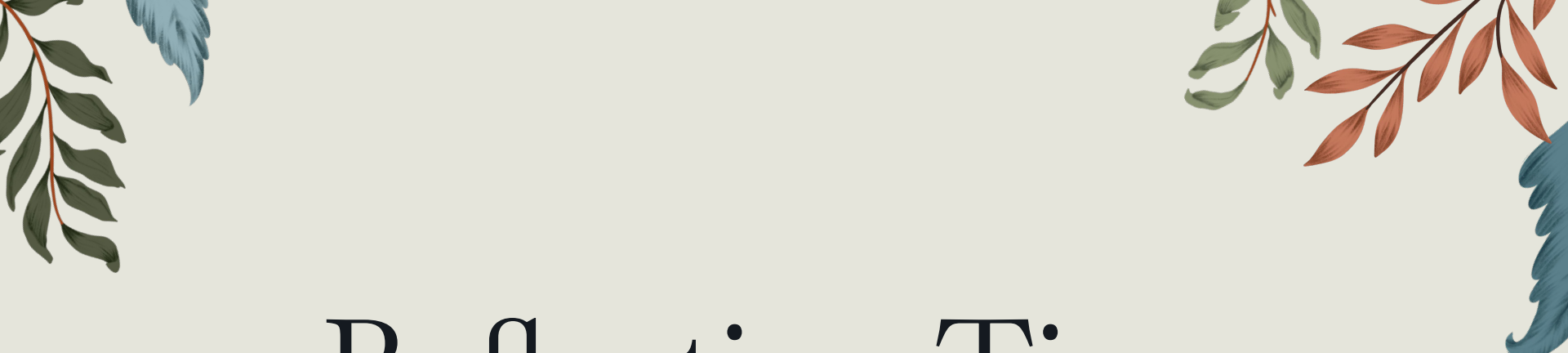
Dialectical

Multiple truths or perspective can exist and be valid at the same time.

Connected

Stay connected with your Posse, your family, and your friends



The top corners of the slide are decorated with stylized floral illustrations. The top-left corner features a branch with dark green leaves and a small blue flower. The top-right corner features a branch with reddish-brown leaves and a small blue flower.

Reflection Time





Discuss 1-2 reflections that inspire you

Reflection 1

There are so many ways to get involved at the intersection of music and sustainability that I didn't even know existed. There's so much to learn, it's exciting and daunting at the same time.

Reflection 2

All this research and work could have not been done without the constant love from my support system. They helped me get through the hard parts and they celebrated with me in successes.





Thanks!

DO YOU HAVE ANY QUESTIONS?

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@anishohmyghosh

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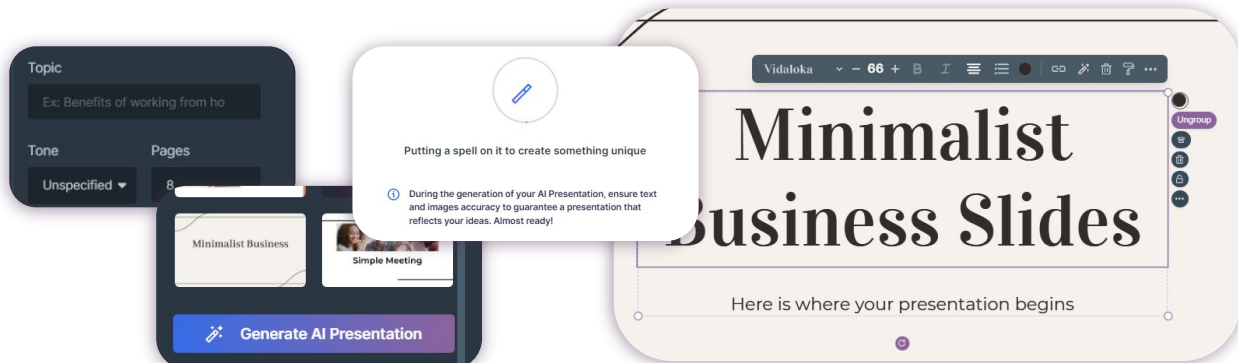
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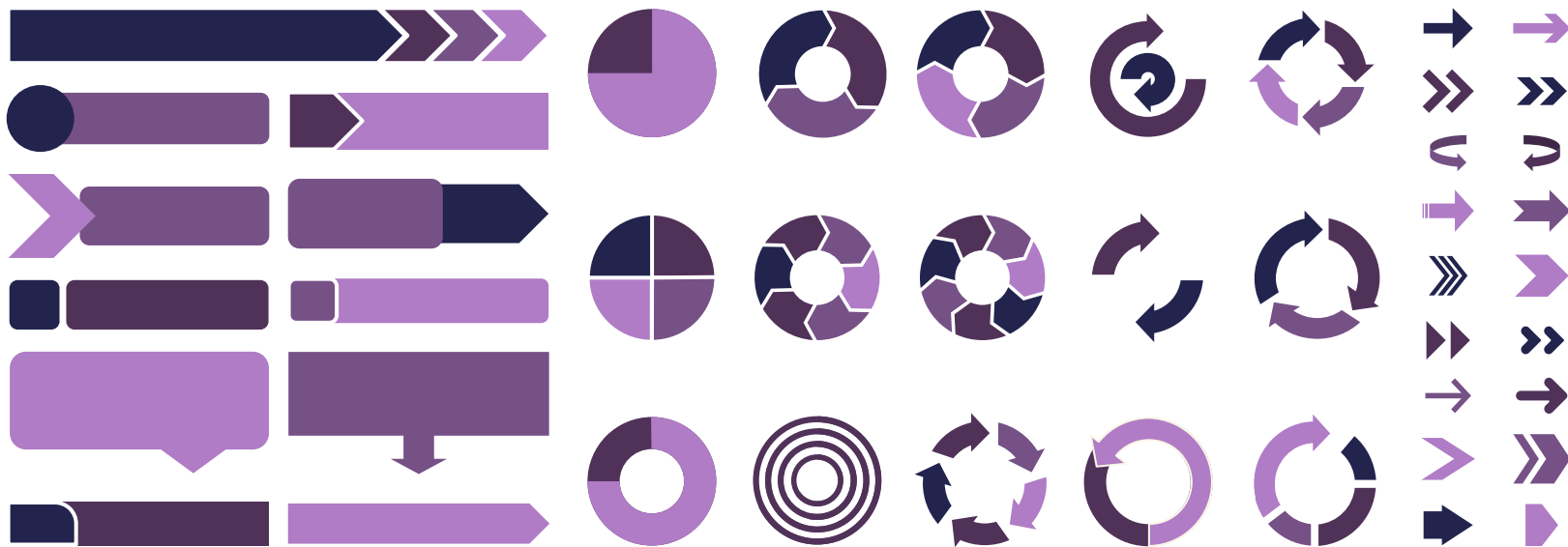
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[Generate AI Presentation](#)

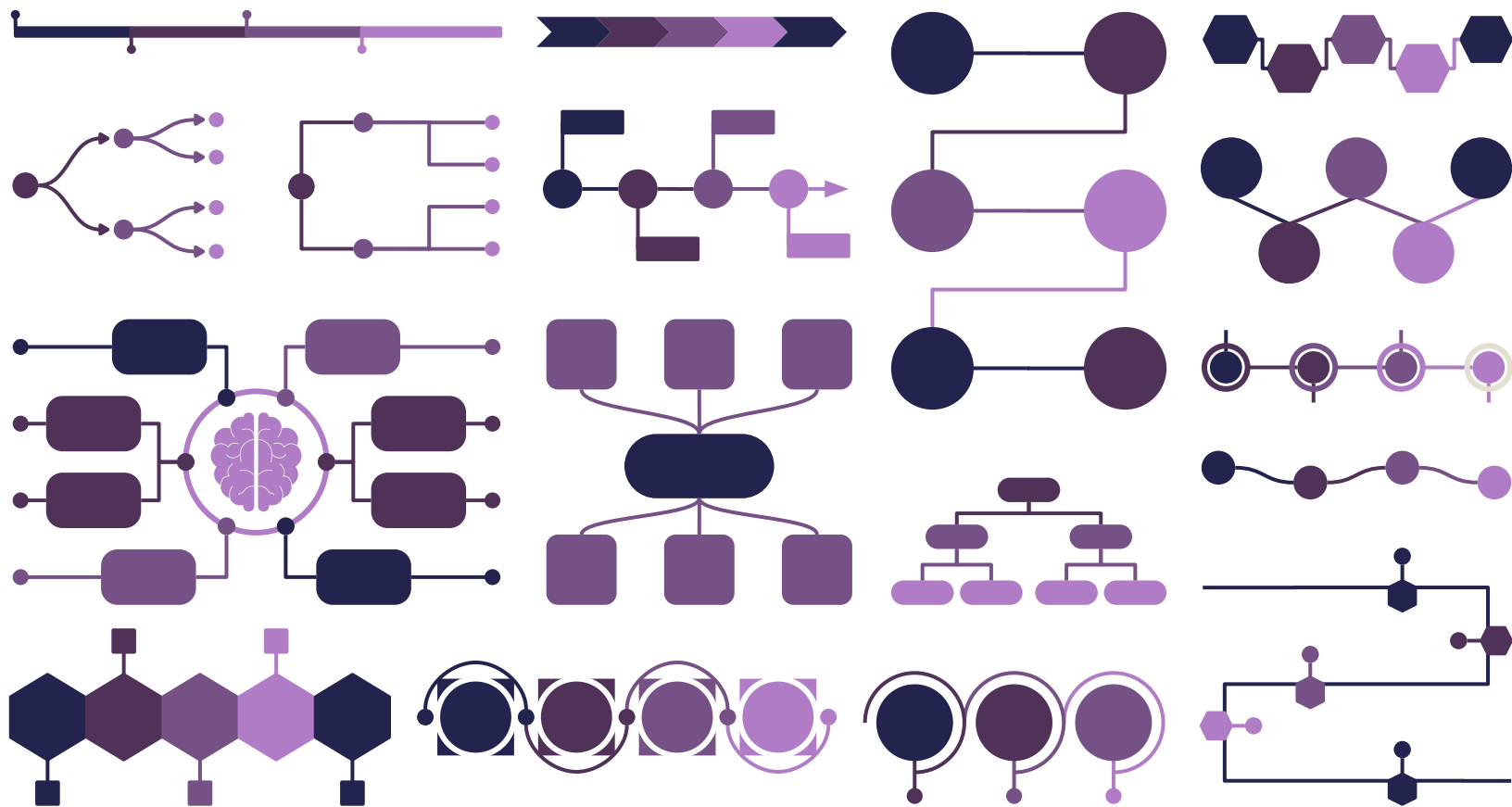
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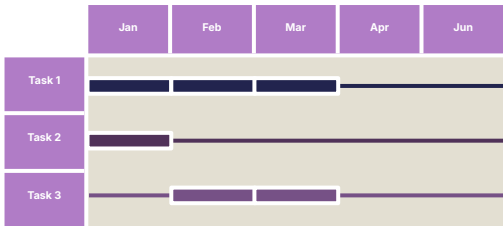
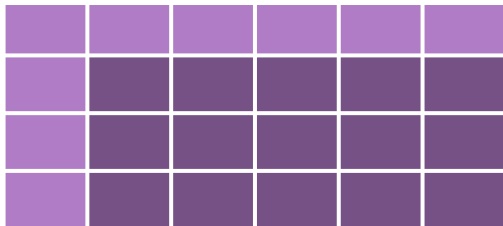
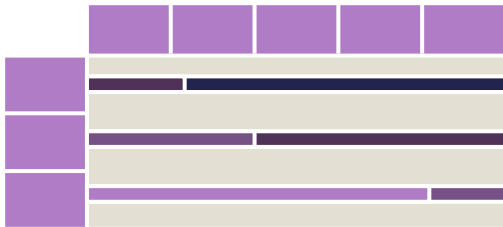
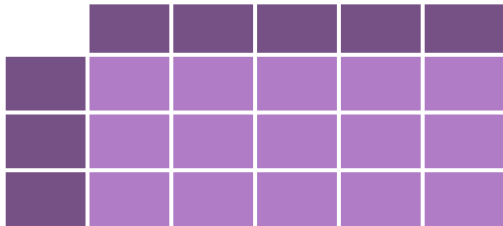
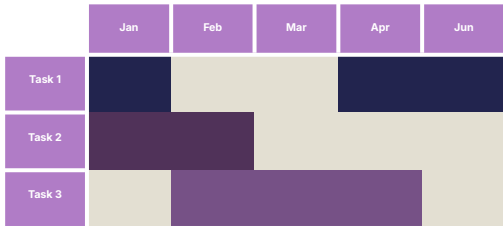
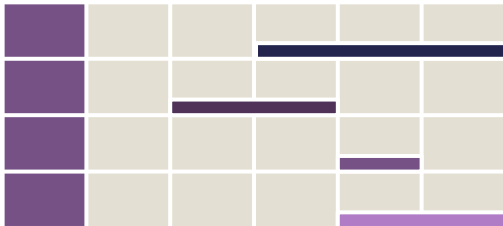
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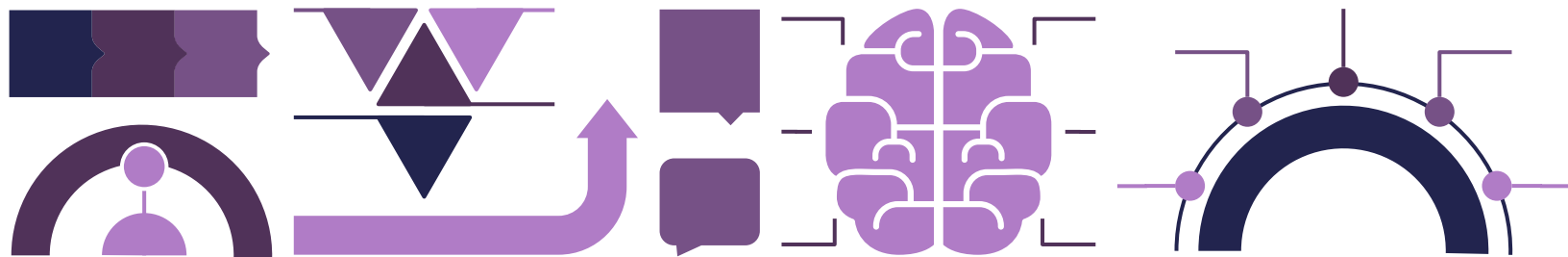
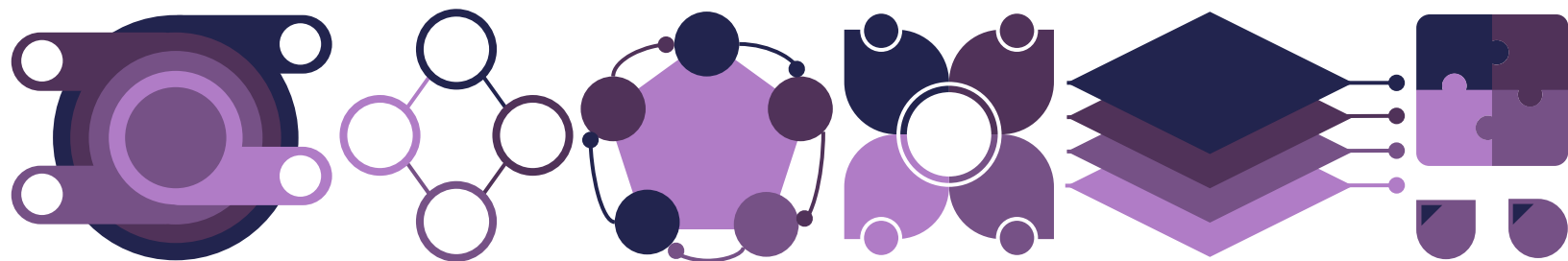
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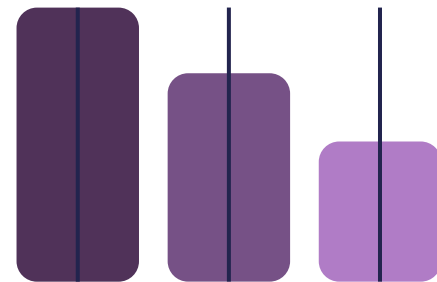
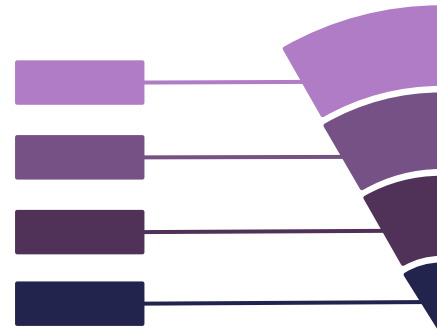
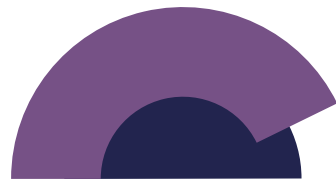
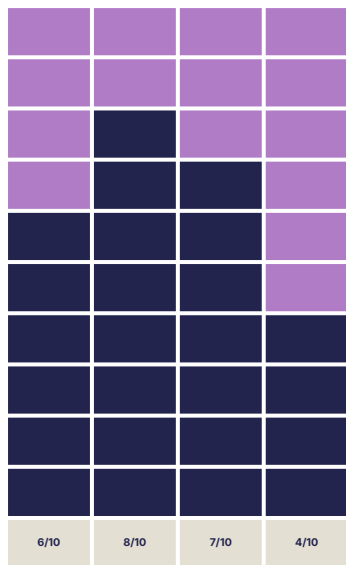












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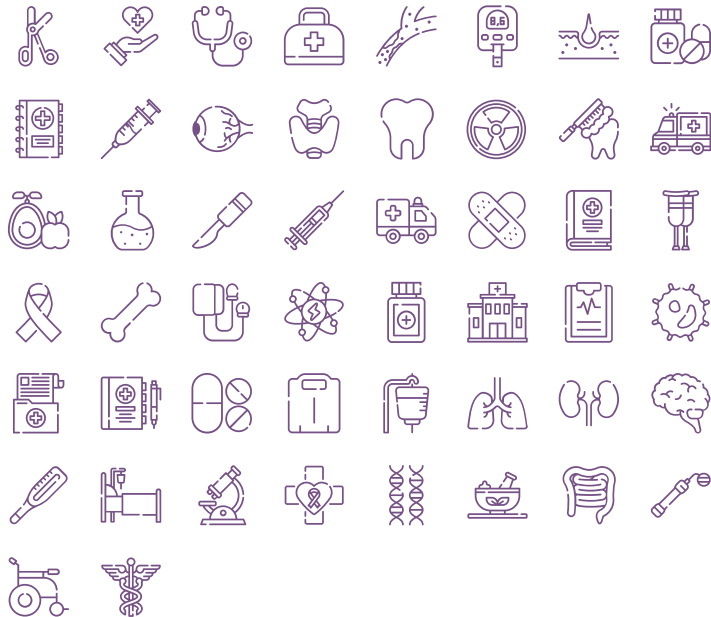
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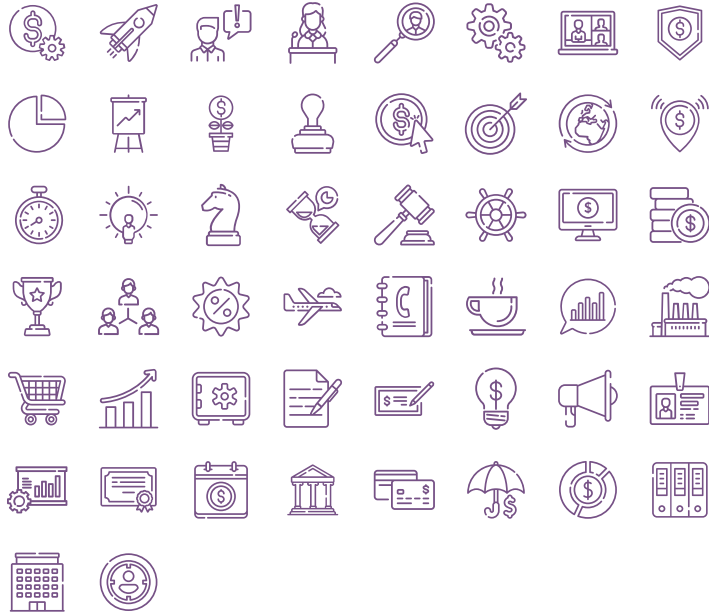
Educational icons



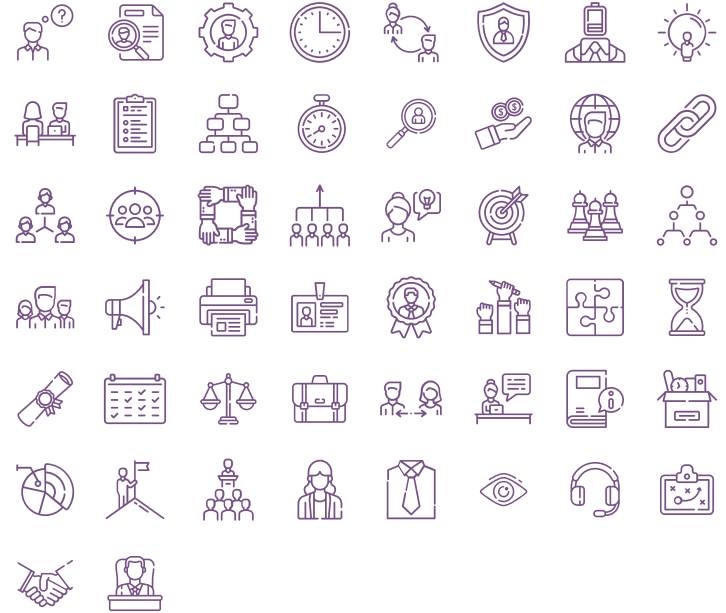
Medical icons



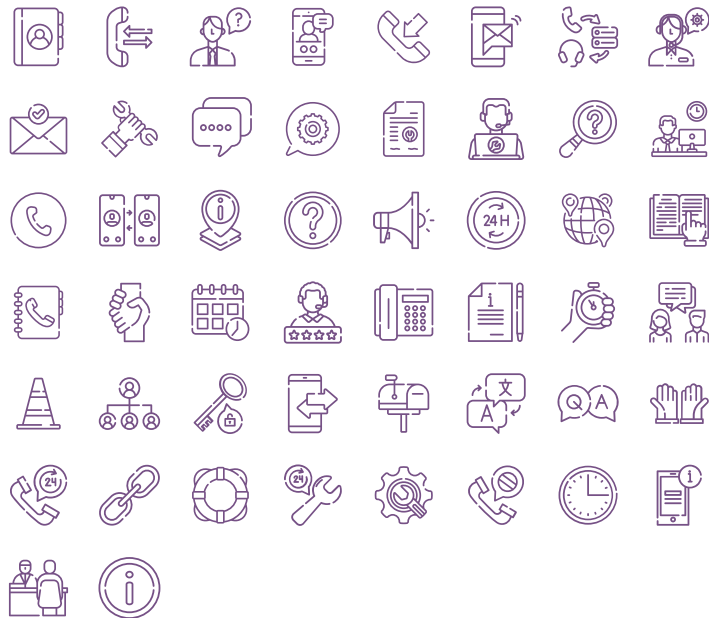
Business icons



Teamwork icons



Help & support icons



Avatar icons



Creative process icons



Performing arts icons



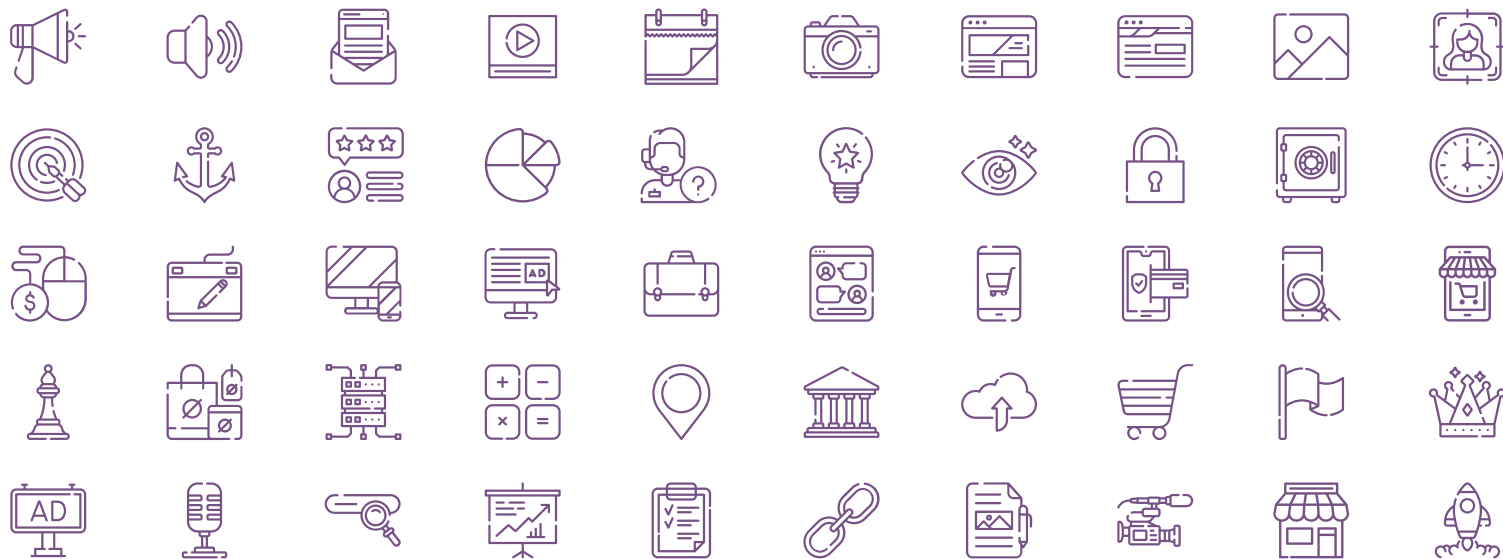
SEO & marketing icons





Premium icons

Digital marketing





Premium icons

Online learning





Premium icons

Laboratory





Premium icons

Goals & results





Premium icons

Infographic elements



